

VIKAASH T

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EDUCATION

B. Tech – Information Technology

2023 – 2027

Velammal College of Engineering and Technology Madurai, Tamil Nadu

■ CGPA – 8.28 (Cumulative up to 6th semester)

Higher Secondary Certificate (HSC)

2022 – 2023

Annamalaiyaar Matric Higher Secondary School Madurai, Tamil Nadu

■ Percentage – 93%

TECHNICAL SKILLS

Programming:	Java, Python
Web Stack:	HTML, CSS, Basic JavaScript
Database:	SQL
ML/DL Frameworks:	Scikit-learn, PyTorch,
Models & Architectures:	Bert, LSTM, T5
Backend:	Flask
Tools:	VS Code, Jupyter Notebook

PROJECTS

Energy Consumption Anomaly Detection System

Python, Flask, Scikit-learn, Random Forest, Pandas

- Built an anomaly detection system for 200 buildings processing **1.4M+ hourly meter readings**, achieving a mean **F1 score of 0.96** across all properties.
- Engineered 15 temporal features including lag variables and rolling statistics; implemented per-building MinMaxScaler fitted exclusively on normal training data.
- Optimized threshold from 0.50 to 0.20, improving anomaly recall from 0.68 to 0.89; selected Random Forest (macro F1 0.92) over XGBoost, LightGBM, LSTM, and Transformer baselines.
- Deployed a clean Flask web app featuring CSV bulk upload infrastructure, processing pipelines, and a data visualization summary dashboard.

IT Ticket Classification System

Python, Flask, BERT, PyTorch, NLP

- Fine-tuned an NLP transformer BERT model to classify IT support tickets into 8 distinct categories, securing **88% accuracy** and a **macro F1 score of 0.88** on 9,500+ textual descriptions.
- Outperformed standard baseline implementations including Logistic Regression (0.85) and Random Forest (0.83) via deep contextual text parsing.
- Developed a Flask web application that processes raw ticket descriptions for single and bulk CSV prediction, with an interactive analytics dashboard displaying category distribution charts and trends.

RESEARCH PUBLICATION

- **Vikaash T, Elavarasi A, Sarveshwar S.** "A Multimodal Explainable Customer Churn Prediction Framework Using Late Fusion and Counterfactual Explanation and Reasoning." *2025 IEEE 7th International Conference on Computing, Communication and Automation (ICCCA)*, 2025.

INTERNSHIP

Machine Learning Intern, Elysium Technology Private Limited

09.12.24 – 28.12.24

- Handled production machine learning lifecycle phases, ranging from initial multi-source raw pipeline cleaning configurations to model deployment evaluation states.
- Constructed a functional classification system designed to track customer attrition metrics, allowing operational teams to proactively spot churn indicators.
- Validated multi-class architectures utilizing Logistic Regression, Random Forests, and XGBoost models; monitored accuracy profiles with explicit ROC-AUC tracking analytics.

AREAS OF INTEREST & SOFT SKILLS

Focus Fields:	Machine Learning, Deep Learning, Natural Language Processing
Core Attributes:	Complex Problem-Solving, Team Collaboration, Technical Documentation